The Closure Residual and Nonlinear Quantum Structure

Equilibrium Quantum Limit and Modulus-Curvature Nonlinearity

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Abstract

Paper XVIII derived an exact nonlinear closure-Nelson wave equation and showed that, at the Paper XIV equilibrium and to leading order near equilibrium, the dynamics reduces to the Schrödinger equation with Witten Hamiltonian H_{W}. Paper XIX isolated the remaining nonlinear term as the closure residual. The present paper places that residual into nonlinear quantum structure.

Writing the dynamics in the equilibrium-centred form

$$i\partial_t \Psi = H_{\mathrm{W}}\Psi + \mathcal{N}_{\mathrm{cl}}[\Psi],$$

the nonlinear operator $\mathcal{N}_{\mathrm{cl}}$ is shown to be a local modulus-curvature term with equilibrium subtraction, fixed entirely by the closure geometry. Its structural properties are derived: phase covariance, amplitude homogeneity, locality away from nodal sets, exact vanishing on the equilibrium amplitude, and perturbative smallness near equilibrium. These results identify the linear Schrödinger equation not as the full closure dynamics, but as the distinguished equilibrium tangent limit of an exact nonlinear wave equation.

The closure nonlinearity is then compared structurally with known classes of nonlinear Schrödinger equations, including modulus-dependent curvature corrections. The claim is not exact identification with any existing family, but a precise mathematical placement: the closure residual defines a derived modulus-curvature nonlinear Schrödinger structure whose coefficient and equilibrium subtraction are fixed by the closure geometry.

This paper does not assign physical interpretation to the nonlinear residual. It establishes that the closure framework yields a mathematically well-defined nonlinear quantum extension with an exact equilibrium quantum limit.

1 Introduction

The closure programme has now constructed a layered route from classical gradient-flow dynamics to quantum-type evolution:

- Paper XVI [Smart, 2026b]: closure evolution equation and structural gap to Schrödinger,
- Paper XVII [Smart, 2026c]: dissipative obstruction for single-generator approaches,
- Paper XVIII [Smart, 2026d]: Nelson pairing and near-equilibrium Schrödinger recovery,
- Paper XIX [Smart, 2026a]: isolation and classification of the nonlinear residual.

Paper XIX established that the residual is a robust, non-trivial, probability-conserving nonlinear remainder that cannot be removed by standard gauge transformations. The purpose of the present paper is not to re-derive that remainder, but to place it within the broader structure of nonlinear Schrödinger dynamics and identify the linear quantum regime as its equilibrium tangent limit.

Purpose

This paper:

1. defines the closure nonlinear operator $\mathcal{N}_{\mathrm{cl}}$ and establishes its structural properties,
2. proves that the linear Schrödinger equation is the equilibrium tangent limit of the full nonlinear dynamics,
3. decomposes the nonlinearity into a universal modulus-curvature term and an equilibrium-fixed subtraction,
4. compares the closure nonlinearity with known nonlinear Schrödinger classes.

It does not claim experimental predictions or exact identification with existing nonlinear QM models. It is a structural classification paper.

2 The Closure Nonlinear Operator and Equilibrium-Centred Form

2.1 Definition

Starting from the Paper XIX normal form, define the *closure nonlinear operator*:

Definition 1 (Closure nonlinear operator)**.**

$$\mathcal{N}_{\mathrm{cl}}[\Psi] = \delta U_{\mathrm{rel}}[\Psi]\,\Psi = \sigma^2 \left(\frac{\Delta|\Psi|}{|\Psi|} - \frac{\Delta R_{\mathrm{st}}}{R_{\mathrm{st}}} \right) \Psi \tag{1}$$

where $R_{\mathrm{st}} = \sqrt{\rho_{\mathrm{st}}} \propto e^{-\mathcal{F}/\sigma^2}$ *and* $\Delta R_{\mathrm{st}}/R_{\mathrm{st}} = 2V_{\mathrm{W}}/\sigma^2$.

The full closure-Nelson dynamics is then:

$$i\,\partial_t \Psi = H_{\mathrm{W}}\Psi + \mathcal{N}_{\mathrm{cl}}[\Psi] \tag{2}$$

2.2 Decomposition

The operator splits into two pieces:

$$\mathcal{N}_{\mathrm{cl}}[\Psi] = \underbrace{\sigma^2 \frac{\Delta|\Psi|}{|\Psi|}\,\Psi}_{\text{universal modulus-curvature term}} - \underbrace{2V_{\mathrm{W}}\,\Psi}_{\text{equilibrium subtraction}} \tag{3}$$

The first term depends only on the modulus profile of Ψ and is the same functional form that appears in the quantum potential of the Madelung formulation. The second term is entirely fixed by the closure equilibrium geometry: it subtracts the equilibrium curvature so that $\mathcal{N}_{\mathrm{cl}}$ vanishes on Ψ_{st}.

Remark 2. *The equilibrium subtraction is not a free parameter. It is determined by $\mathcal{F}$ and σ through $\Delta R_{\mathrm{st}}/R_{\mathrm{st}} = 2V_{\mathrm{W}}/\sigma^2$. This is the key structural difference between the closure nonlinearity and a generic nonlinear Schrödinger term: the subtraction is derived from the closure landscape, not introduced phenomenologically.*

3 Structural Properties

Proposition 3 (Phase covariance)**.** *For any constant $\theta \in \mathbb{R}$:*

$$\mathcal{N}_{\mathrm{cl}}[e^{i\theta}\Psi] = e^{i\theta}\mathcal{N}_{\mathrm{cl}}[\Psi] \tag{4}$$

That is, $\mathcal{N}_{\mathrm{cl}}$ is covariant under global phase rotations.

Proof. $|e^{i\theta}\Psi| = |\Psi|$, so $\delta U_{\mathrm{rel}}[e^{i\theta}\Psi] = \delta U_{\mathrm{rel}}[\Psi]$, and $\mathcal{N}_{\mathrm{cl}}[e^{i\theta}\Psi] = \delta U_{\mathrm{rel}}[\Psi] \cdot e^{i\theta}\Psi = e^{i\theta}\mathcal{N}_{\mathrm{cl}}[\Psi]$. □

Proposition 4 (Amplitude homogeneity)**.** *For any constant $\lambda > 0$:*

$$\mathcal{N}_{\mathrm{cl}}[\lambda\Psi] = \lambda\,\mathcal{N}_{\mathrm{cl}}[\Psi] \tag{5}$$

Proof. $|\lambda\Psi| = \lambda|\Psi|$, so $\Delta|\lambda\Psi|/|\lambda\Psi| = \Delta|\Psi|/|\Psi|$. Then $\delta U_{\mathrm{rel}}[\lambda\Psi] = \delta U_{\mathrm{rel}}[\Psi]$, and $\mathcal{N}_{\mathrm{cl}}[\lambda\Psi] = \delta U_{\mathrm{rel}}[\Psi] \cdot \lambda\Psi = \lambda\mathcal{N}_{\mathrm{cl}}[\Psi]$. □

Corollary 5 (Combined scaling)**.** $\mathcal{N}_{\mathrm{cl}}[\lambda e^{i\theta}\Psi] = \lambda e^{i\theta}\mathcal{N}_{\mathrm{cl}}[\Psi]$ *for $\lambda > 0$, $\theta \in \mathbb{R}$. The operator is homogeneous of degree one and phase-covariant.*

Proof. Immediate from Propositions 3 and 4. □

Proposition 6 (Locality)**.** *On any open set where $|\Psi| > 0$, $\mathcal{N}_{\mathrm{cl}}[\Psi](\psi)$ depends only on $|\Psi|$, $\nabla|\Psi|$, $\Delta|\Psi|$, Ψ, and the equilibrium data R_{st} evaluated at ψ. It is a local nonlinear operator.*

Proof. Immediate from the definition (1): $\Delta|\Psi|/|\Psi|$ involves only the values and first two spatial derivatives of $|\Psi|$ at the point ψ, and $\Delta R_{\mathrm{st}}/R_{\mathrm{st}}$ is a fixed function of ψ. □

Proposition 7 (Equilibrium vanishing)**.**

$$\mathcal{N}_{\mathrm{cl}}[\Psi_{\mathrm{st}}] = 0 \tag{6}$$

The equilibrium state lies in the kernel of the nonlinear operator. On the equilibrium amplitude sector, the dynamics is exactly linear Schrödinger.

Proof. Direct: $|\Psi_{\rm st}| = R_{\rm st}$, so $\Delta|\Psi_{\rm st}|/|\Psi_{\rm st}| = \Delta R_{\rm st}/R_{\rm st}$ and the difference in (1) vanishes. □

Proposition 8 (Modulus-only dependence)**.** *$\mathcal{N}_{\rm cl}[\Psi]$ depends on the phase of Ψ only through the overall factor Ψ in (1). The residual potential $\delta U_{\rm rel}[\Psi]$ is a functional of $|\Psi|$ alone.*

Proof. The quantity $\Delta|\Psi|/|\Psi|$ depends only on the modulus and its derivatives, not on the phase of Ψ. The equilibrium term $\Delta R_{\rm st}/R_{\rm st}$ is phase-independent. Thus $\delta U_{\rm rel}[\Psi]$ is modulus-only; the factor Ψ in $\mathcal{N}_{\rm cl}[\Psi] = \delta U_{\rm rel}[\Psi]\Psi$ carries the phase. □

4 The Equilibrium Quantum Limit

The linear Schrödinger equation is not imposed on the closure framework; it emerges as the distinguished tangent dynamics at equilibrium.

Proposition 9 (Equilibrium tangent limit)**.** *Let $\Psi(t) = \Psi_{\rm st}(1 + \varepsilon\,\eta(\psi, t))$ with $\varepsilon \ll 1$. Then the Fréchet linearisation of the full nonlinear flow (2) at the equilibrium state $\Psi_{\rm st}$ is the Witten-Hamiltonian Schrödinger generator: to leading order,*

$$i\,\partial_t(\Psi_{\rm st}\,\eta) = H_{\rm W}(\Psi_{\rm st}\,\eta) + O(\varepsilon) \tag{7}$$

That is, infinitesimal perturbations around equilibrium evolve under the linear Schrödinger equation $i\partial_t\delta\Psi = H_{\rm W}\,\delta\Psi$.

Proof. Since $\mathcal{N}_{\rm cl}[\Psi_{\rm st}] = 0$ (Proposition 7) and $\delta U_{\rm rel}$ is continuous in $|\Psi|$, the residual potential is Fréchet-small under $\Psi = \Psi_{\rm st}(1 + \varepsilon\eta)$: $\delta U_{\rm rel}[\Psi] = O(\varepsilon)$ (Paper XIX, Proposition 4). The nonlinear contribution is of higher perturbative order than the linear Witten term, so the Fréchet linearisation of the full flow is generated by $H_{\rm W}$. □

Remark 10 (Interpretation)**.** *The Witten-Hamiltonian Schrödinger equation $i\partial_t\Psi = H_{\rm W}\Psi$ is not the full closure dynamics. It is the tangent linearisation of the full nonlinear dynamics at the equilibrium fixed point $\Psi_{\rm st}$. Linear quantum mechanics is the equilibrium sector of the closure theory, in the same sense that linear stability analysis is the equilibrium sector of a nonlinear dynamical system.*

This is a structural result: the linear quantum regime is selected by the equilibrium geometry, not by postulate.

5 Relation to the Madelung Quantum Potential

In the Madelung formulation of standard quantum mechanics, the Hamilton–Jacobi equation contains the *quantum potential*:

$$Q_{\rm std} = -\frac{\hbar^2}{2m}\frac{\Delta\sqrt{\rho}}{\sqrt{\rho}} \tag{8}$$

The closure nonlinearity involves the same functional form:

$$\mathcal{N}_{\rm cl}[\Psi] = \sigma^2\frac{\Delta|\Psi|}{|\Psi|}\,\Psi - 2V_{\rm W}\,\Psi \tag{9}$$

The first term is proportional to $(\Delta\sqrt{\rho}/\sqrt{\rho})\Psi$ with coefficient σ^2 (in Schrödinger-normalised form; σ^4 in the undivided Paper XVIII form). The structural connection is:

Table 1: Quantum potential in standard QM vs. the closure nonlinearity.

Feature	Standard QM	Closure framework
Functional form	$\Delta\sqrt{\rho}/\sqrt{\rho}$	$\Delta\lvert\Psi\rvert/\lvert\Psi\rvert$ (same)
Coefficient	$-\hbar^2/(2m)$	$+\sigma^2$ (normalised)
Sign in HJ equation	Negative	Positive (reversed)
Equilibrium subtraction	None	$-2\sigma^2 V_{\mathrm{W}}$ (closure-derived)
Role in wave equation	Absorbed into linear kinetic term	Residual nonlinear correction

Remark 11. *In standard QM, the quantum potential is entirely absorbed into the linear Schrödinger kinetic term — it is not a separate nonlinear correction. In the closure framework, the sign reversal (Paper XVIII) prevents full absorption, leaving a residual nonlinear term. The equilibrium subtraction then ensures this residual vanishes at the closure ground state. The closure residual is therefore not merely "another nonlinear term"; it is the non-cancelled remainder of the same modulus-curvature object that, in standard linear quantum mechanics, is hidden inside the Schrödinger kinetic structure.*

6 Relation to Nonlinear Schrödinger Classes

The closure equation (2) belongs to the broad class of nonlinear Schrödinger equations of the form:

$$i\,\partial_t\Psi = \hat{H}_0\Psi + f(|\Psi|, \nabla|\Psi|, \Delta|\Psi|)\,\Psi \tag{10}$$

where $\hat{H}_0$ is a linear Hamiltonian and f is a real-valued nonlinear functional of the modulus.

6.1 Modulus-curvature nonlinearities

The closure nonlinearity has the specific form:

$$f_{\mathrm{cl}} = \sigma^2\frac{\Delta|\Psi|}{|\Psi|} - 2V_{\mathrm{W}} \tag{11}$$

This is a *modulus-curvature* nonlinearity: it depends on the Laplacian of the modulus (second-order spatial derivatives), not on powers of the modulus. This distinguishes it from:

- **Gross–Pitaevskii** ($f = g|\Psi|^2$): power-law, no derivatives.
- **Logarithmic** ($f = \lambda\log|\Psi|^2$): no derivatives, but singular.

It is structurally closest to:

- **Doebner–Goldin-type** nonlinearities, which include terms proportional to $\Delta|\Psi|/|\Psi|$ and arise from representations of the diffeomorphism group on quantum states.

6.2 Comparison with generic modulus-curvature NLS

A generic modulus-curvature nonlinear Schrödinger equation has the form:

$$i\partial_t \Psi = \hat{H}_0 \Psi + D \frac{\Delta|\Psi|}{|\Psi|} \Psi \tag{12}$$

with D a real coupling constant. The closure equation differs in two ways:

1. **The coefficient is fixed.** In the closure framework, $D = \sigma^2$ (in Schrödinger-normalised form) is determined by the noise scale, not a free phenomenological parameter.

2. **An equilibrium subtraction is present.** The closure equation subtracts $2\sigma^2 V_{\mathrm{W}}$, which ensures the nonlinearity vanishes at the closure equilibrium. A generic modulus-curvature NLS has no such subtraction.

Remark 12 (Structural placement)**.** *The closure equation is not presented as identical to the Doebner–Goldin family or any specific nonlinear QM model. It may be placed within the same broad class of modulus-curvature nonlinear Schrödinger equations, but with two closure-specific features: a derived coefficient (σ^2 in normalised form) and a derived equilibrium subtraction ($2V_{\mathrm{W}}$). These features are consequences of the closure geometry, not free parameters.*

Remark 13 (On relation to nonlinear quantum theories)**.** *The closure equation is not proposed as an alternative nonlinear quantum theory. Its role here is structural: to show that a nonlinear Schrödinger-type equation with a distinguished linear limit arises naturally from the closure construction. Whether this structure corresponds to a physical nonlinear extension of quantum mechanics is a separate question not addressed in this paper.*

7 What the Closure Framework Fixes

In a generic nonlinear Schrödinger theory of the form (10), the nonlinear functional f is typically a phenomenological input. The closure framework fixes several features that would otherwise be free:

Table 2: What the closure framework determines vs. what is free in generic NLS.

Feature	Generic NLS	Closure framework
Linear Hamiltonian	Free choice	H_{W} (Witten, from $\mathcal{F}$, σ)
Nonlinear functional form	Free choice	$\Delta\|\Psi\|/\|\Psi\|$ (modulus curvature)
Nonlinear coefficient	Free (D)	Fixed (σ^2, normalised)
Equilibrium subtraction	Absent	Present ($2V_{\mathrm{W}}$, from $\mathcal{F}$)
Distinguished linear limit	Not automatic	Equilibrium tangent dynamics
Norm conservation	Must be checked	Guaranteed (real δU_{rel})

Within the forward/backward closure-pairing construction, the framework therefore produces a *more constrained* nonlinear Schrödinger structure than a generic phenomenological model: fewer free parameters, a distinguished linear limit, and a derived equilibrium subtraction.

8 Worked Example: Quadratic Closure

Example 14 (Quadratic closure)**.** *Let $\mathcal{F}(\psi) = \frac{1}{2}k\psi^2$ in one dimension.*

Closure data

$R_{\mathrm{st}} \propto e^{-k\psi^2/(2\sigma^2)}$, $V_{\mathrm{W}} = k^2\psi^2/(2\sigma^2) - k/2$, H_{W} *has spectrum* $E_n = nk$.

Nonlinear operator on specific states

Ground state $\Psi = \phi_0 \propto R_{\mathrm{st}}$*:*

$$\mathcal{N}_{\mathrm{cl}}[\phi_0] = 0 \qquad \textit{(exact Schrödinger sector)} \tag{13}$$

Shifted Gaussian $|\Psi| \propto e^{-k(\psi-a)^2/(2\sigma^2)}$ *with displacement* a*:*

$$\delta U_{\mathrm{rel}} = k^2(-2a\psi + a^2) \tag{14}$$

The residual potential is linear in ψ (plus constant), growing with displacement a. For small a, $|\delta U_{\mathrm{rel}}| \ll |V_{\mathrm{W}}|$ and the dynamics is in the Schrödinger regime.

First excited state $|\Psi| = |\phi_1| \propto |\psi|\, e^{-k\psi^2/(2\sigma^2)}$*:*

$$\delta U_{\mathrm{rel}} = 2\sigma^2/\psi^2 - 2k \qquad (\psi \neq 0) \tag{15}$$

The residual diverges at the node $\psi = 0$ (where $|\Psi| = 0$) and is large near the origin. The excited eigenstate dynamics is in the nonlinear closure regime near nodes.

Regime classification

Table 3: Residual strength for different states in the quadratic model.

State	δU_{rel}	Ξ	Regime
Ground ϕ_0	0	0	Exact Schrödinger
Shifted Gaussian ($a \ll \sigma/k$)	$O(a)$	$\ll 1$	Linear Schrödinger
Shifted Gaussian ($a \gg \sigma/k$)	$O(k^2a^2)$	~ 1 or $\gg 1$	Mixed / nonlinear
Excited ϕ_1 (away from node)	$O(\sigma^2 k)$	$O(1)$	Mixed
Excited ϕ_1 (near node)	Divergent	$\gg 1$	Nonlinear closure

The quadratic example demonstrates the full regime structure: exact Schrödinger at the ground state, perturbative Schrödinger for small displacements, and genuinely nonlinear dynamics for large excitations or near nodal sets. Accordingly, statements involving excited states should be understood piecewise on nodal domains unless a regularised treatment is supplied.

9 Interpretation and Limits

What is established

The closure residual defines a specific nonlinear Schrödinger structure:

- a modulus-curvature nonlinearity with derived coefficient σ^2 (in Schrödinger-normalised form),
- an equilibrium subtraction fixed by the closure geometry,
- a distinguished linear quantum limit at equilibrium,
- norm conservation in all regimes.

What is not claimed

- **No exact identification with Doebner–Goldin or any specific NLS family.** The closure equation belongs to the same broad class but has closure-specific features (equilibrium subtraction, derived coefficient) not present in generic models.
- **No experimental prediction.** Determining whether the nonlinear residual produces observable effects requires estimating $\sigma^2\Delta|\Psi|/|\Psi|$ in physically realistic regimes, which is not attempted here.
- **No claim about the physical status of the residual.** Whether it represents genuine beyond-QM physics, an effective-theory artefact, or a gauge-removable structure remains open.

The structural message

The closure framework does not terminate at the linear quantum regime. It produces a nonlinear Schrödinger structure whose exact linear limit is the equilibrium-centred Witten-Hamiltonian regime. The nonlinearity is not a defect; it is a structural prediction of the paired-process construction, with a coefficient and subtraction term determined by the closure geometry.

Whether this prediction is physical is a question for subsequent work.

10 Discussion

10.1 Exact results

- The closure nonlinear operator $\mathcal{N}_{\mathrm{cl}}$ is defined and its decomposition into modulus-curvature and equilibrium-subtraction terms is exact (Section 2).
- Phase covariance, amplitude homogeneity, locality, and equilibrium vanishing are proved (Section 3).
- The equilibrium tangent limit is proved: linear Schrödinger dynamics is the linearisation of the full nonlinear dynamics at equilibrium (Section 4).
- The quadratic example is computed explicitly (Section 8).

10.2 Structural observations

- The closure nonlinearity belongs to the modulus-curvature class of nonlinear Schrödinger equations (Section 6).
- The coefficient σ^2 (normalised) and equilibrium subtraction $2V_{\mathrm{W}}$ are closure-derived, not phenomenological (Section 7).
- The quantum potential of the Madelung formulation and the closure residual share the same functional form but differ in sign and role (Section 5).

10.3 Open questions

- Does the closure nonlinear equation have a variational structure or a conserved energy functional?
- Can the nonlinear regime be constrained by existing experimental bounds on nonlinear quantum mechanics?
- Is the closure equation exactly equivalent to a specific Doebner–Goldin subfamily, or only structurally related?
- Does the nodal divergence of the residual indicate a physical boundary condition or a need for regularisation?

11 Conclusion

Paper XX places the closure residual into nonlinear quantum structure.

Main result. The closure-Nelson dynamics is a derived nonlinear Schrödinger equation of modulus-curvature type, with coefficient σ^2 (in Schrödinger-normalised form) and equilibrium subtraction $2V_{\mathrm{W}}$ both fixed by the closure geometry. Linear quantum mechanics is not the full closure dynamics; it is the distinguished equilibrium tangent limit.

What the closure framework provides. Compared with a generic nonlinear Schrödinger model, the closure framework fixes: the linear Hamiltonian (Witten), the nonlinear functional form (modulus curvature), the nonlinear coefficient (σ^2 in normalised form), the equilibrium subtraction ($2V_{\mathrm{W}}$), and the existence of a distinguished linear limit. These features are derived from the closure functional $\mathcal{F}$ and the noise scale σ, not postulated.

Programme arc.

- Papers XII–XIV: dissipative closure dynamics,
- Paper XV: phase and interference,
- Paper XVI: evolution equation and structural gap,
- Paper XVII: dissipative obstruction,
- Paper XVIII: Nelson pairing and near-equilibrium Schrödinger recovery,
- Paper XIX: isolation and classification of the closure residual,

- Paper XX: structural placement as a derived nonlinear Schrödinger equation with equilibrium quantum limit.

The next stage is to determine whether the nonlinear residual is experimentally constrainable, whether it predicts observable deviations from standard quantum mechanics, and whether the closure framework thereby extends, modifies, or (in the equilibrium limit) exactly recovers the quantum-mechanical description of nature.

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